Supplementary Material

Appendix: Perimetric Outcomes of Melbourne Rapid Field Perimetry in Patients with Glaucoma: A Systematic Review and Meta-Analysis

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GITHUB LINK: Complete access to all supplementary materials, including the PROSPERO registration, original protocol, extracted dataset, and full statistical analysis code, is provided at the following link: https://github.com/andresinzunza/pop_tag

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Search Strategy

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Date: March 11, 2025

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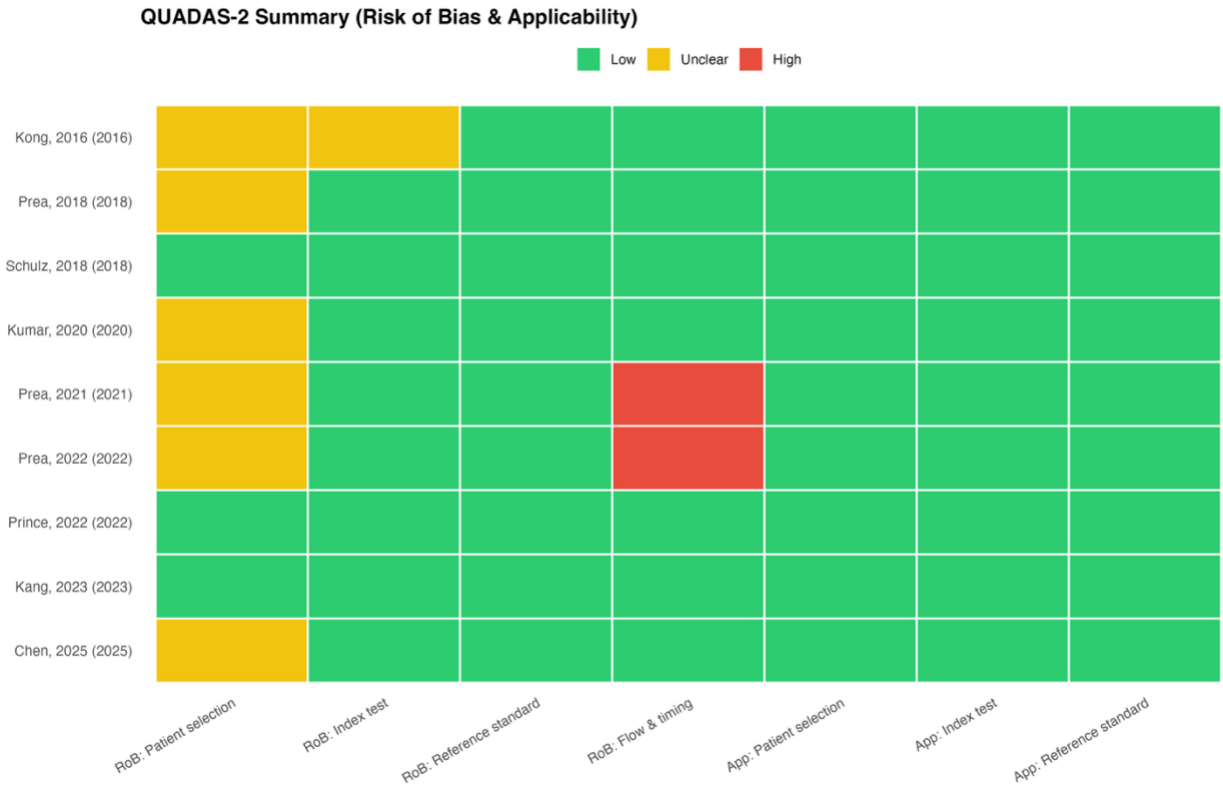
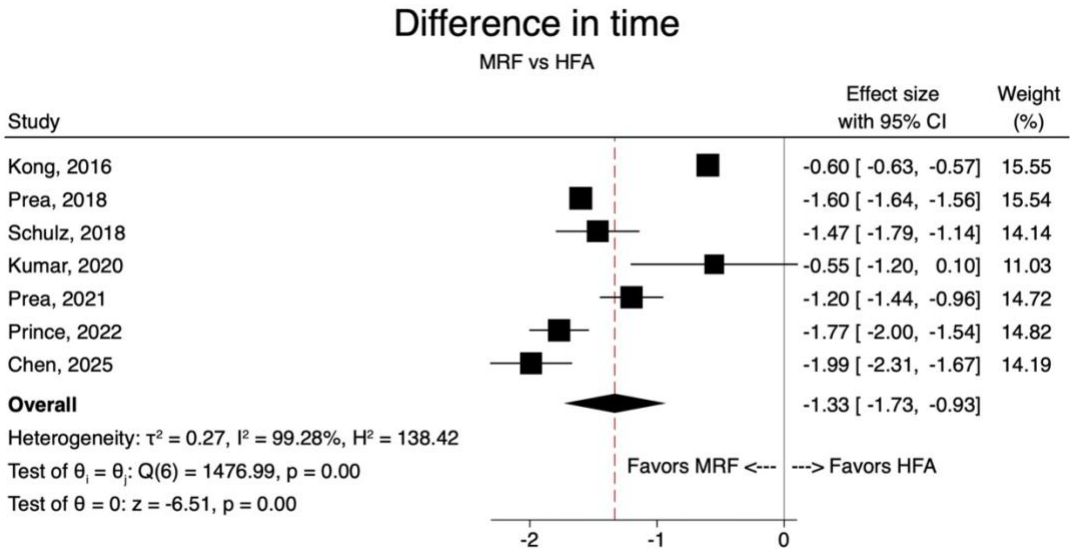
Risk of Bias

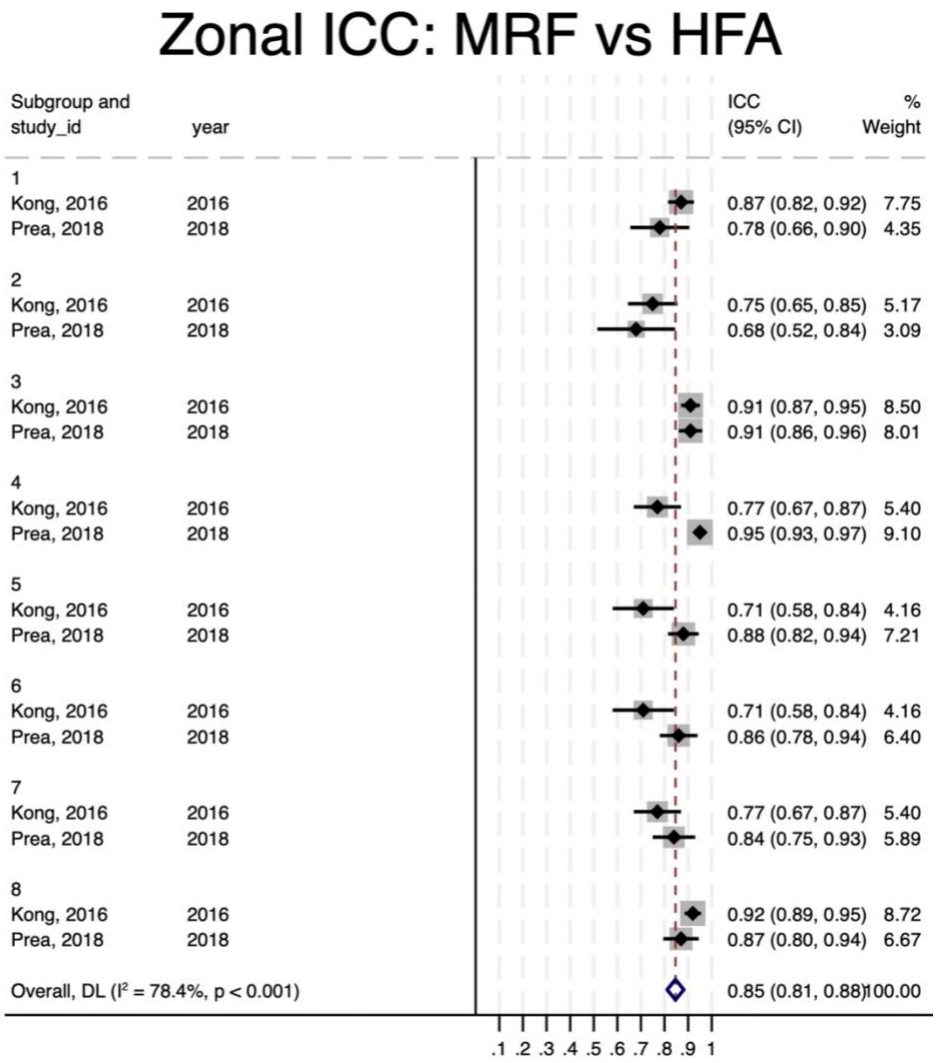
Evaluator: Andrés Inzunza, MD (Mass Eye and Ear/Mass General Brigham)
Date: January 2026



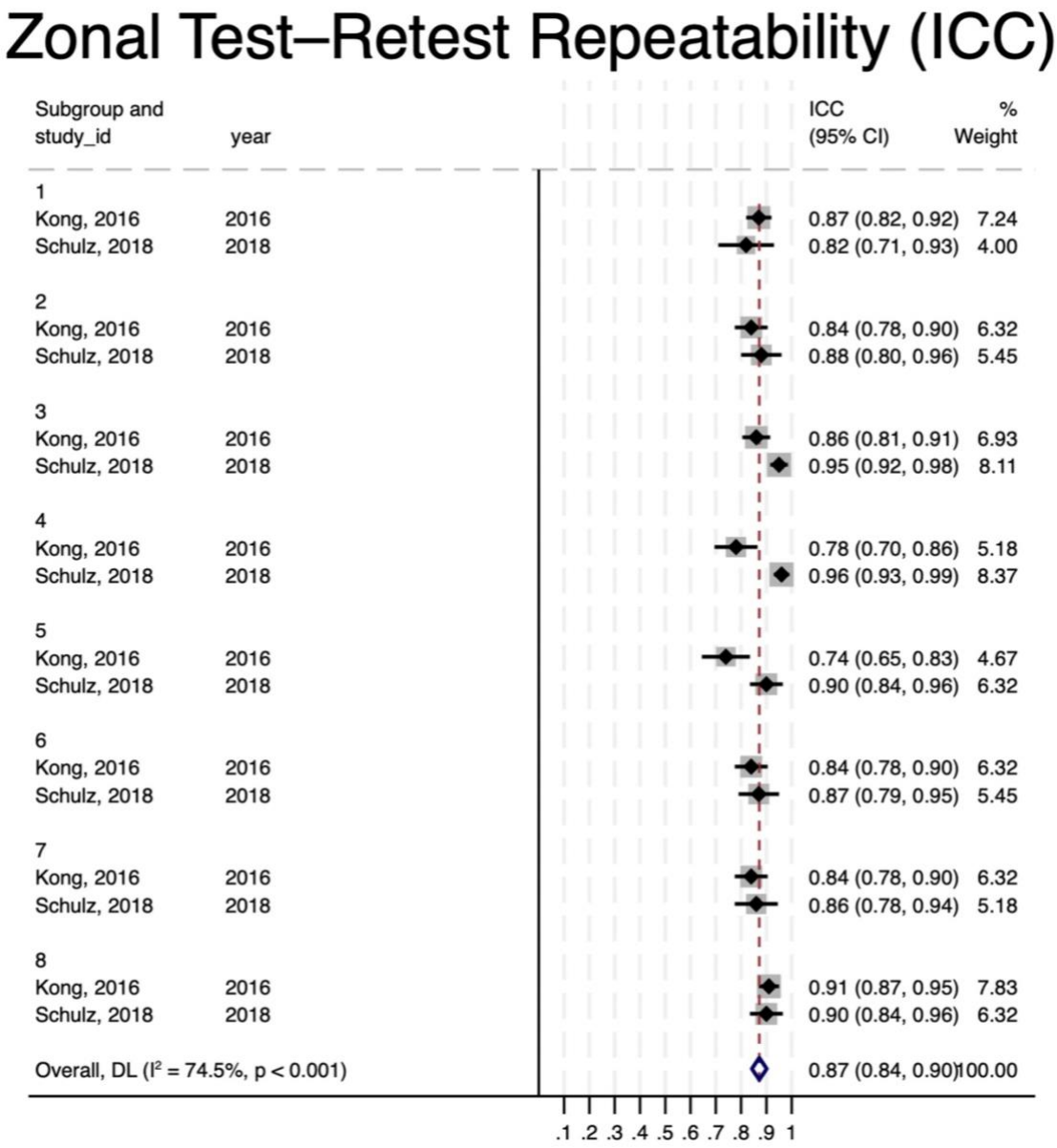
Supplementary Figures



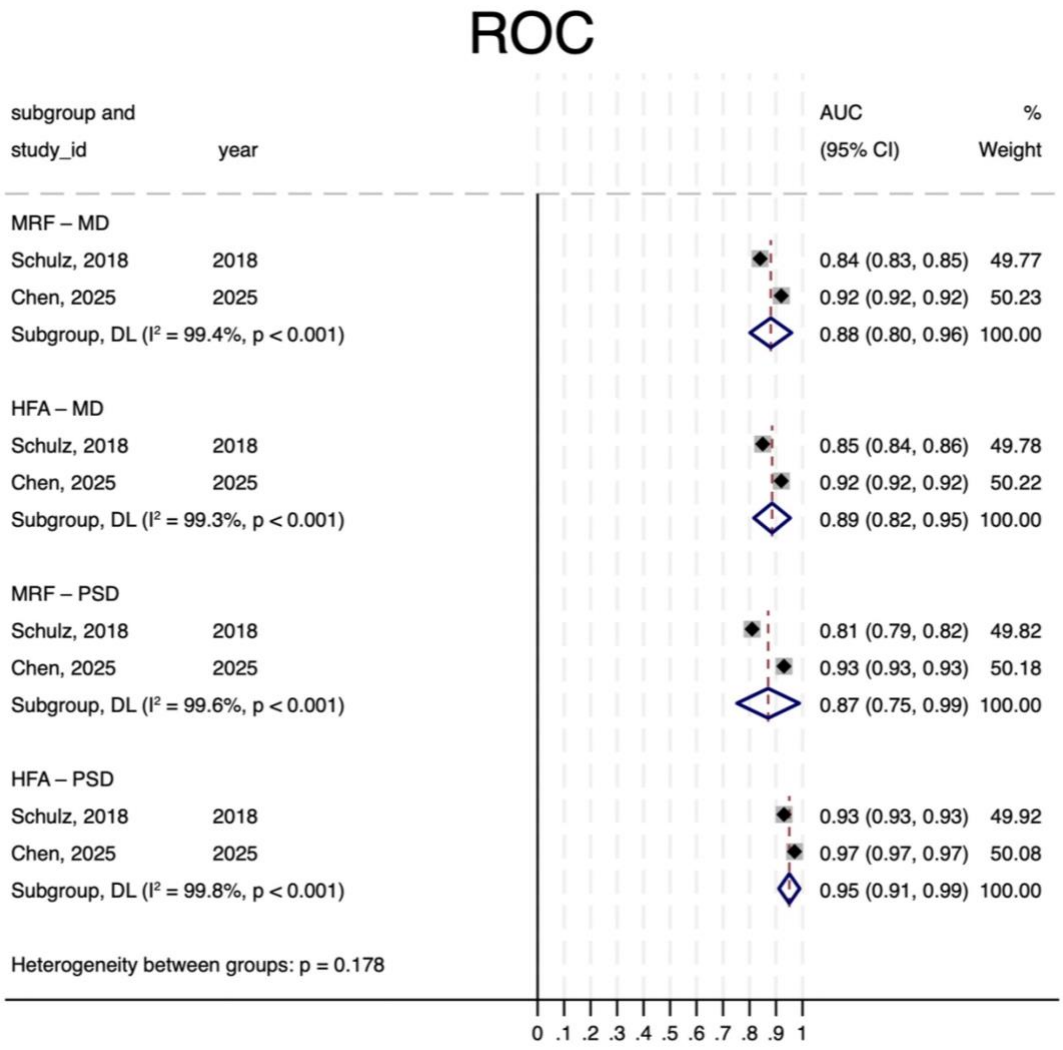
Supplementary Figure 1. Forest plot comparing per-eye testing time between Melbourne Rapid Fields (MRF) and Humphrey Field Analyzer (HFA). The pooled mean difference indicates that MRF testing is approximately 1.3 minutes (approximately 1 minute and 20 seconds) shorter per eye than HFA. Negative values favor MRF.



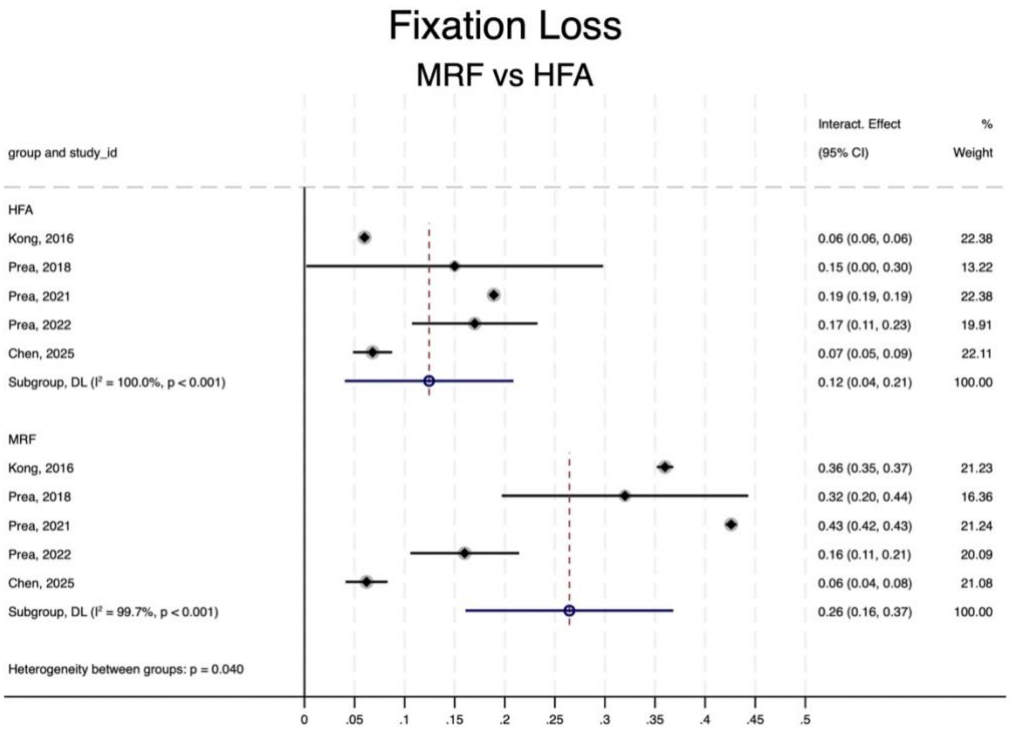
Supplementary Figure 2. Forest plot of zonal intraclass correlation coefficients (ICCs) comparing regional mean threshold values between Melbourne Rapid Fields (MRF) and Humphrey Field Analyzer (HFA). Pooled ICCs are shown for each of the eight visual field zones: Zone 1 (superior–central), Zone 2 (inferior–central), Zone 3 (superior–peripheral–nasal), Zone 4 (superior–nasal), Zone 5 (superior–temporal), Zone 6 (inferior–temporal), Zone 7 (inferior–nasal), and Zone 8 (inferior–peripheral–nasal), as well as the overall pooled estimate across zones. Higher ICC values indicate stronger concordance between MRF and HFA regional threshold measurements.



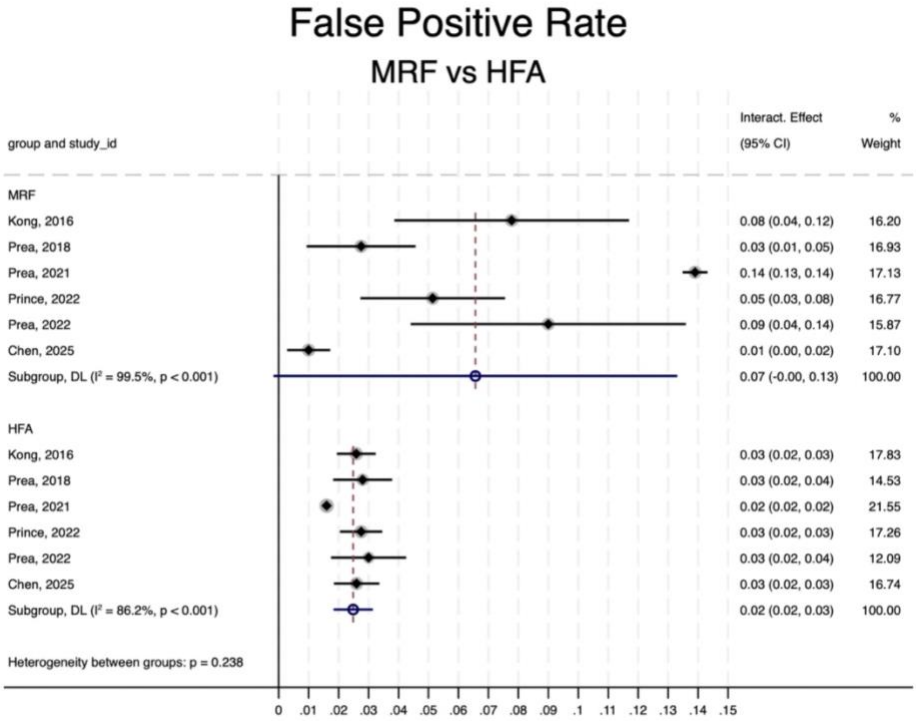
Supplementary Figure 3. Forest plot of zonal intraclass correlation coefficients (ICCs) for test–retest repeatability of regional mean threshold values obtained with Melbourne Rapid Fields (MRF). Pooled ICCs are shown for each of the eight visual field zones: Zone 1 (superior–central), Zone 2 (inferior–central), Zone 3 (superior–peripheral–nasal), Zone 4 (superior–nasal), Zone 5 (superior–temporal), Zone 6 (inferior–temporal), Zone 7 (inferior–nasal), and Zone 8 (inferior–peripheral–nasal), as well as the overall pooled estimate across zones. Higher ICC values indicate greater test–retest repeatability of MRF regional threshold measurements.



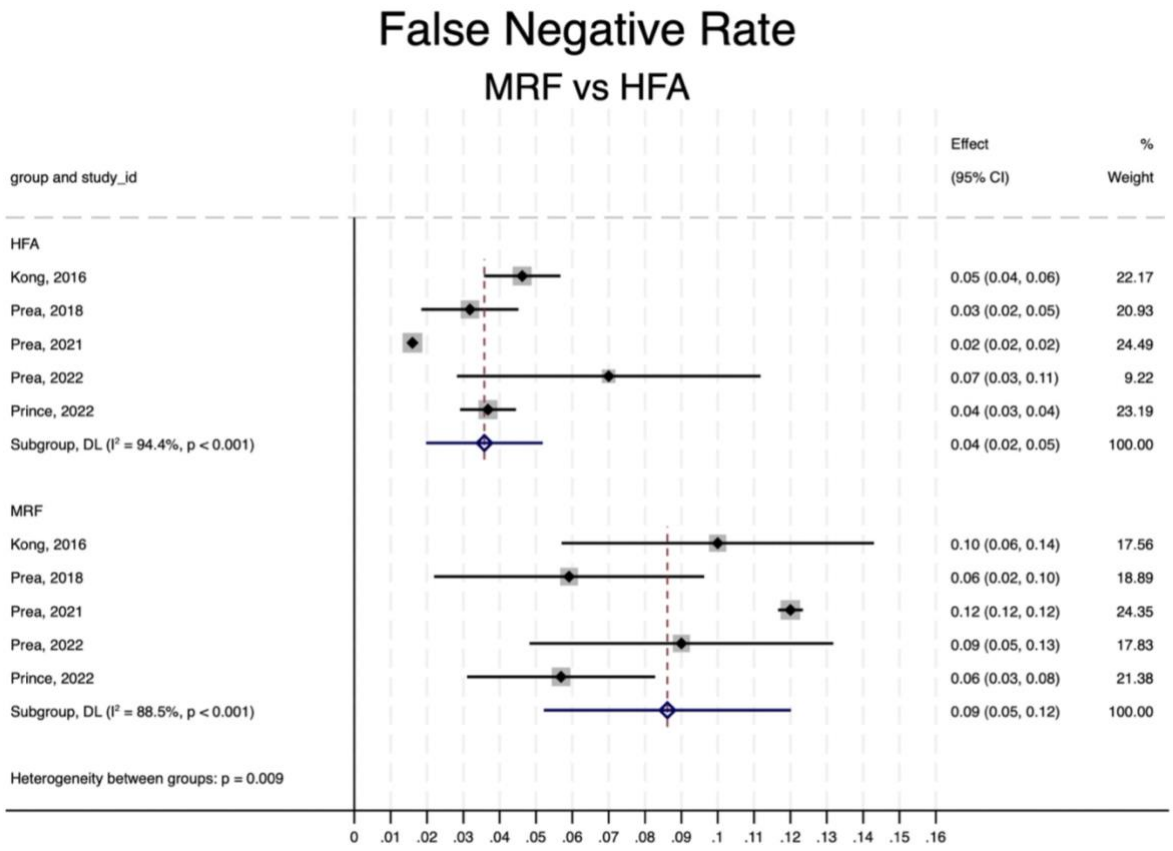
Supplementary Figure 4. Forest plot of pooled areas under the receiver operating characteristic curve (AUCs) for the discrimination of glaucomatous versus normal eyes using mean deviation (MD) and pattern standard deviation (PSD) derived from Melbourne Rapid Fields (MRF) and Humphrey Field Analyzer (HFA). Subgroup analyses are shown for MRF MD, HFA MD, MRF PSD, and HFA PSD. Higher AUC values indicate better discriminatory performance. **Note**: Normal subjects were classified as such by an ophthalmologist following an ophthalmologic examination



Supplementary Figure 5. Forest plot of the pooled proportion (raw proportions) of fixation losses for Humphrey Field Analyzer (HFA) and Melbourne Rapid Fields (MRF). The sub-group comparison assesses differences in fixation loss proportions between devices. Higher proportions indicate a greater frequency of fixation losses.



Supplementary Figure 6. Forest plot of the pooled proportion of false-positive responses for Melbourne Rapid Fields (MRF) and Humphrey Field Analyzer (HFA). Of note, the pooled confidence interval for the MRF subgroup is markedly wide, reflecting substantial between-study heterogeneity and limited precision in this estimation.



Supplementary Figure 7. Forest plot of the pooled proportion of false-negative responses for Melbourne Rapid Fields (MRF) and Humphrey Field Analyzer (HFA). The pooled false-negative proportion is higher for MRF than for HFA.